INCH-POUND

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MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS, AN/AIC-14A
MICROPHONE MIXING MODULE

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Microphone Mixing Module described herein shall consist of this specification and the latest issue of MIL-I-85846.

REQUIREMENTS:

Item description: The Microphone Mixing Module shall provide amplification of two independent audio signals and filtered microphone current as specified herein. The amplifier(s) shall combine the signals to form a single output.

Item diagram: The Microphone Mixing Module shall provide the functions represented schematically in accordance with the diagram of Figure 1.

Interface description: The Microphone Mixing Module input and output signals will normally display the characteristics presented in Table I.

Environmental: The Microphone Mixing Module shall meet the environmental requirements of MIL-I-85846 except the fluids requirement shall not apply.

Characteristics:

Performance characteristics:

Input characteristics:

Microphone No. 1 (MIC NO. 1): The Microphone Mixing Module shall have a NO. 1 input circuit having the following performance characteristics:

Frequency and Amplitude: The MIC NO. 1 shall accept input drive signals which range from 300 to 7000 Hz in frequency and 0.01 to 1.0 Vac in amplitude. The module shall be capable of operating within specifications when the signal input contains a DC component of up to 2.0 volts.

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TABLE I. Signal characteristics.

Signal Name	Characteristics
Inputs:	
Microphone No. 1 (MIC NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1.0 Vac. No dc component normally present.
Microphone No. 2 (MIC NO. 2)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1.0 Vac. No dc component normally present.
Power (PWR)	+27.5 ± 0.5 Vdc.
Outputs:	
Audio (AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 2.0 Vac.
Microphone Current (MIC CUR)	55 ± 10 milliamperes dc (into a 490 ohm load).
Grounds:	
Power (PWR GND)	Power return. See MIL-I-85846.
System (SYS GND)	Signal return. See MIL-I-85846.

Input impedance: The MIC NO. 1 input impedance shall be 100 ohms ± 20 percent.

Microphone No. 2 (MIC NO. 2): The MIC NO. 2 inputs shall be the same as those for MIC NO. 1.

Power (PWR): The Microphone Mixing Module shall accept and operate within the constraints of this specification when supplied with electrical power as specified in MIL-I-85846. The maximum load presented by the Microphone Mixing Module shall be not greater than 150 milliamperes.

The Microphone Mixing Module shall contain an insertion filter for the MIC CUR output which shall be capable of attenuating ripple on the +27.5 Vdc line a minimum of 20 dB within the range of 300 to 7000 Hz.

Output characteristics: The Microphone Mixing Module shall have the following output characteristics:

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Microphone Audio (MIC AUDIO): The MIC AUDIO output shall have the following characteristics:

Frequency and amplitude: The AUDIO signal output amplitude shall equal the input amplitude at 1000 Hz within ±2 dB. Within the frequency range from 300 to 7000 Hz the signal amplitude shall not vary more than 2 dB above or below the value at 1000 Hz when terminated in a 100 ohm resistive load. The input/output characteristic (gain/loss) of each amplifier shall be unity (0 dB) and not vary more than ±2 dB when the input is varied from 0.01 to 1.0 volts at 1000 Hz.

Distortion: The total harmonic distortion over the frequency range of 300 to 7000 Hz over the amplitude range of 0.01 to 1.0 Vac shall be not greater than five percent under standard conditions as specified in MIL-I-85846. The total harmonic distortion over the frequency range from 300 to 7000 Hz shall be not greater than eight percent throughout the environmental envelope of MIL-E-5400 Class 2 as modified in MIL-I-85846.

Output power: Each amplifier shall be capable of delivering a minimum of 14 milliwatts into a 100 ohm non-inductive load.

Amplifier stability: The output signal shall be stable and show no evidence of regeneration or oscillation when operated with input and output circuits terminated in the specified impedance, open circuit, short circuit, and/or any combination thereof. No damage shall result from amplifier operation with inputs or outputs terminated in either open or short circuits.

Internal noise: The amplifier shall have an output of less than 5 millivolts throughout the environmental envelope in MIL-E-5400 for Class 2 equipment as modified by MIL-I-85846 with the input circuit shorted to system ground.

Susceptibility to Stray Magnetic Fields: The module shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 to 2800 Hz, the output from the amplifier shall be not greater than 5 millivolts at 400 Hz. Between 400 and 2800 Hz the output shall be not greater than a linear increase up to 50 millivolts at 2800 Hz.

Microphone Current (MIC CUR): The MIC CUR output shall provide 55 ± 10 ma of current to a 490 ohm load.

Physical characteristics:

Dimensions: The Microphone Mixing Module shall consist of a single encapsulated module having the dimensions of Figure 2. All outside dimensions shall be within ±0.03 inch of the nominal value shown.

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Weight: Weight of the Microphone Mixing Module shall be not greater than 3.0 ounces.

Design and construction: The design and construction of the Microphone Mixing Module shall be in accordance with the requirements of MIL-I-85846.

QUALITY ASSURANCE PROVISIONS: See MIL-I-85846.

Responsibility for tests and inspections: See MIL-I-85846.

Test methods:

Examination of product: See MIL-I-85846. Units procured as spares shall not be subjected to the fluids inspection (see 4.2.1.10 of MIL-I-85846).

Operational test: See MIL-I-85846.

Input impedance: Verify that the NO. 1 and NO. 2 meets the requirements herein.

Gain: Verify that the amplifier gain at 1000 Hz meets the requirements herein.

Frequency Response: Holding the input signal constant vary the frequency between 300 Hz and 7000 Hz. Verify that the amplifier gain meets the requirements herein.

Linearity: Using a 1000 Hz input signal which is varied between 0.01 Vac and 1.0 Vac, verify that the amplifier meets the requirements herein.

Output power: Verify the output requirements contained herein, by applying a 1000 Hz signal of appropriate amplitude and terminating the Audio Out pin with a pull up network. (The network shall consist of a 100 ohm resistor in series with a 390 ohm resistor which is connected to a +27.5 $\pm$ .5 Vdc power source. The junction of the two resistors shall be shunted to ground thru a 80 microfarad capacitor. The Audio Out pin shall be connected to the remaining lead of the 100 ohm resistor.)

Distortion: Verify the requirements herein.

Amplifier stability: Verify the requirements herein.

Internal noise: Verify the requirements herein.

Susceptibility to Stray Magnetic Fields: Verify the performance herein.

Environmental stress screening (burn-in): All environmental stress screening inspection shall be required on all MIC Mixing Modules. Only those modules that have successfully completed the environmental stress screening inspection shall be delivered to the procuring activity. Prior to and after each submission to the screening inspection, a room ambient full electrical performance inspection shall be performed on each MIC Mixing Module. Each MIC Mixing Module shall be subjected to 10 4-hour

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temperature cycles consisting of a 2-hour cold period at -54°C and a 2-hour hot period at +71°C. The modules need not be energized during the temperature cycling portion of this inspection. The thermal stress screening inspection chamber's temperature change rate shall be greater than 10°C per minute.

Special inspections: See MIL-I-85845.

Preparing Activity:
Navy - AS
(Project No. 5831-N084)

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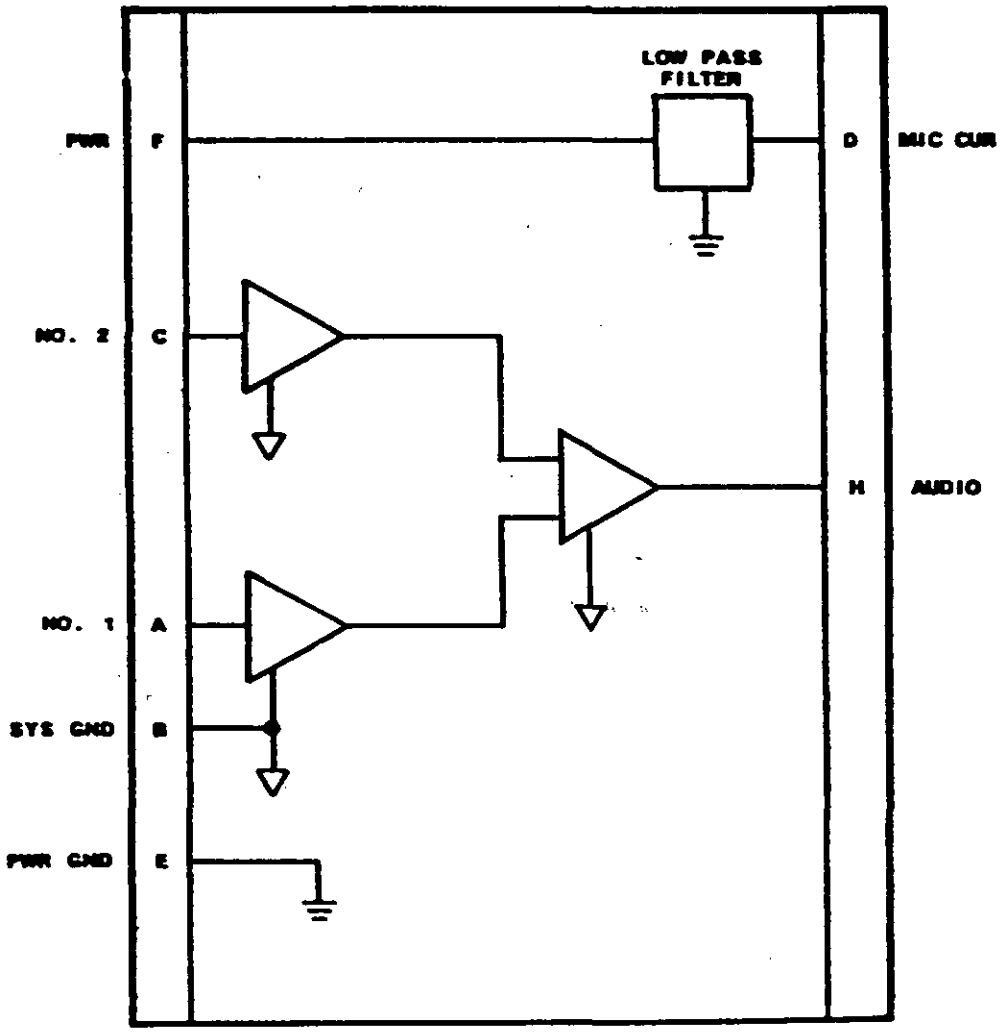
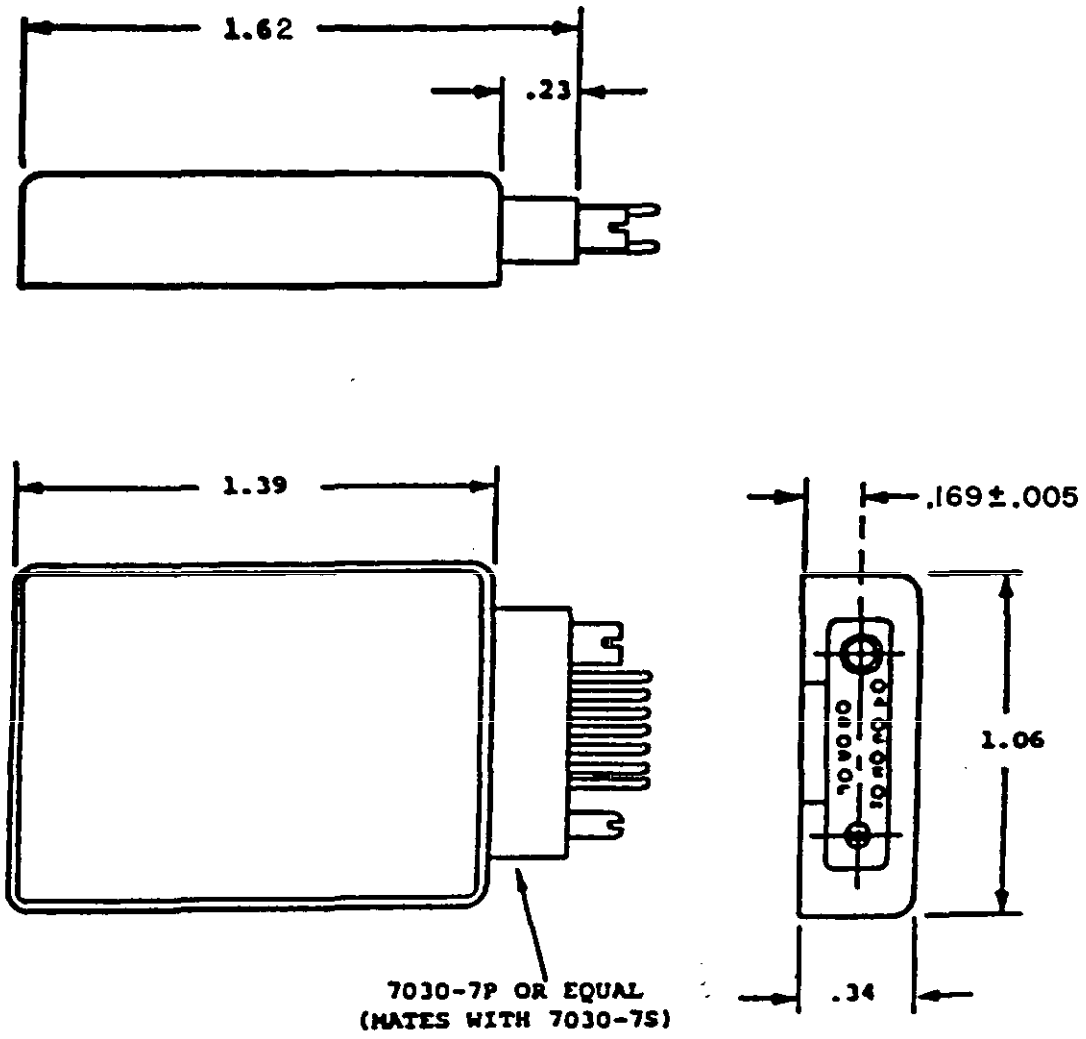


FIGURE 1. Microphone mixing module functional diagram.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.01 inches unless otherwise specified.

FIGURE 2. Microphone mixing module outline drawing.

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